

Question	Marking Guidance	Mark	Comments
04.1	Correct answer of 120 = 3 marks;;; Answers in range 117 to 119 = 2 marks;; 0.1017 - 0.1018 or 0.102 (correct calculation of πr^2) = 1 mark OR 12 divided/multiplied by their incorrectly calculated area = 1 mark;	3 (3 x AO2)	

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04.2	Count the number of stomata in many parts of leaf and calculate a mean/average;	1 (1 x AO1)	Accept multiple/many fields of view/samples Ignore several Ignore multiple leaves/plants If number stated must be 10 or more

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04.3	Any one of: Hairs so 'trap' water vapour and water potential gradient decreased/low OR Stomata in pits/grooves/sunken stomata so 'trap' water vapour/reduce air movement/maintain humidity and water potential gradient decreased/low OR Thick (waxy) cuticle so increases diffusion distance OR (Thick) waxy cuticle so prevents/reduces evaporation/transpiration;	1 (1 x AO1)	Accept: humid/moist air as 'water vapour' but not water/moisture on its own Accept: diffusion/concentration gradient as equivalent to water potential gradient Ignore 'no' water loss

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04.4	1. As wind speed increases the rate of transpiration increases, but at a slower rate; 2. Rate of transpiration/evaporation is higher in the non-xerophyte (than the xerophyte) OR Higher wind speed has a greater effect on the non-xerophyte than the xerophyte; 3. Some transpiration/evaporation occurs (in both types) in still air/no wind;	2 max (2 x AO3)	